

DANIEL ADIF NUGROHO

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Digital Transformation Engineer | Data Engineer | AI/ML Practitioner

PROFESSIONAL SUMMARY

Results-driven Digital Transformation Engineer with 4+ years at PT Mattel Indonesia, specializing in Python-based backend systems, GCP cloud architecture, scalable ETL pipelines, and AI/ML deployment. Proven experience building LLM-powered production assistants, AI-based compliance detection systems on Google Cloud Platform, OEE forecasting models, and Power BI analytics platforms. M.Sc. in Computer Science (Business Intelligence focus) with hands-on expertise in deploying production-grade AI applications, integrating ML models into manufacturing workflows, and driving measurable operational improvements through data-driven systems.

CORE COMPETENCIES

Languages & Frameworks: Python, SQL, VBA | FastAPI, Flask, REST APIs

Cloud & Infrastructure: Google Cloud Platform (GCP) — AI model execution, backend architecture, cloud-native deployment

AI / ML: LLM integration, Prompt Engineering, Predictive Modeling, MLP, Computer Vision, Anomaly Detection, Time-Series Forecasting, Explainable AI (XAI)

Data Engineering: ETL pipeline development, data modeling, AS400, SQL Server, structured/unstructured data processing

BI & Analytics: Power BI, DAX, dashboard design, KPI reporting, NLP on operational text

Tools & Libraries: PyTorch, Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib

Practices: Distributed systems, scalable APIs, OEE optimization, production lifecycle analytics

PROFESSIONAL EXPERIENCE

Digital Transformation Engineer | PT Mattel Indonesia

Apr 2025 – Present

- Designed and built an LLM-powered manufacturing assistant on GCP enabling real-time production decision-making, equipment troubleshooting, process guidance, and operational knowledge retrieval on the shop floor.
- Architected and deployed an AI-powered packaging compliance detection system on Google Cloud Platform — handling artwork ingestion, AI model execution, compliance validation against regulatory documents, and structured reporting — reducing manual review effort and enabling earlier detection of regulatory packaging issues.
- Developing an OEE forecasting engine through machine-product correlation analysis, modeling relationships between machine parameters, product attributes, and production KPIs to estimate and optimize future equipment effectiveness.
- Leading end-to-end delivery of AI/data initiatives aligned to Mattel's digital transformation roadmap across manufacturing operations.

Data Engineer | PT Mattel Indonesia

Aug 2022 – Apr 2025

- Engineered and automated scalable ETL pipelines for production machine data, transforming raw operational data into structured, analysis-ready datasets — reducing report size by 90% and improving Power BI dashboard load speed by 80%.
- Built optimized data models and Power BI dashboards delivering reliable operational reporting, improving manufacturing visibility, and supporting faster business decision-making across production teams.
- Applied NLP techniques to analyze unstructured text from daily gemba reports, identifying recurring production issues, operational patterns, and improvement opportunities for proactive shop-floor decision support.

Data Engineer (Intern) | PT Mattel Indonesia*Dec 2021 – Aug 2022*

- Assisted stakeholders in building and improving Power BI dashboards, translating reporting requirements into clear visualizations that enhanced data accessibility and decision-making.
- Built an Excel VBA-based inventory management system for tooling operations, enabling transaction tracking and reducing manual tracking effort.
- Contributed to ETL process design, collecting and integrating data from AS400, Excel, and SQL sources to support web-based application backends.

Industrial Engineering Intern | PT Mattel Indonesia*Oct 2021 – Dec 2021*

- Supported operational process analysis and KPI reporting, gaining foundational experience in manufacturing workflows and industrial performance metrics.

EDUCATION

Master's Degree, Computer Science | President University*Jan 2023 – Jun 2025*

GPA: 3.81 / 4.00

Thesis: MLP-Driven Prediction of Machine Status for Enhanced Maintenance in Industrial Systems — Applied Multi-Layer Perceptron models to forecast machine conditions from production status logs, enabling proactive predictive maintenance for SMEs.

Bachelor of Engineering, Industrial Engineering | President University*Aug 2018 – Mar 2022*

GPA: 3.45 / 4.00

KEY AI / DATA PROJECTS

- GCP Packaging Compliance AI: End-to-end backend system on Google Cloud Platform for regulatory artwork verification using AI model execution, structured reporting, and automated compliance validation.
- LLM Manufacturing Assistant: GPT-based production assistant enabling real-time shop-floor guidance, equipment troubleshooting, and operational knowledge retrieval.
- OEE Forecasting Model: Machine-product correlation analytics to predict and optimize Overall Equipment Effectiveness using time-series and predictive modeling techniques.
- ETL Pipeline Automation: Scalable pipelines processing raw machine production data — 90% report size reduction, 80% Power BI performance improvement.
- Predictive Maintenance MLP: Research-grade neural network model for machine failure prediction from industrial status log data (M.Sc. thesis).

ADDITIONAL

Teaching: Part-time Lecturer — Data, Analytics & Technology (President University)

Interests: Applied AI in manufacturing, cloud-native ML deployment, digital transformation, business intelligence